



CTP Mathaverse

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0.1 Induction, Comments, Idea, Interesting Stories

CTP is a computer, He was a student in CUHK Mathematics major. He is not good at Mathematics and English. Dear Dr Fong pointed out that he didn't have enough foundation to handle undergraduate mathematics. He felt loss and meaningless about studying too deep mathematics. One day he met his MATH JESSUS Kelvin and good friends (Wiki, Alex, Andy, Ricky, Jason, Yuki, Oscar, Jason), Yuki inspired him to make a notes or source to cover all proof and idea in undergraduate mathematics. Even he will get a Third Hon but he still contributed the society ?.....Here is it the Mathematics Universe by CTP's friends.

0.2 Principal of Math Major

1. DON'T CHEAT, it shows your respect to this hard question.
2. Write Down Definition (you can get 1 mark in CUHK)
3. Prove Directly (this case in wonderland)
4. Prove by Contradiction (if you don't know, try this)
5. Prove by Contrapositive (hope you don't need to use it)
6. Condition False then the whole statement true (so STUBID)
7. Don't cry during mid-term and final (Professor will kill you)
8. KEEP QUIET (especially FINAL and MIDTERM)
9. Be Humble (as there are lot of Asians better than you)
10. Discuss with your friends and holds a study group!
11. you can try DEM BEAT to memorize the math theorem or proof??
12. Suppose bla bla bla, then blo blo blo.
There exists some bla bla bla then blo blo blo
For any bla bla bla then blo blo blo
13. If you want to late-drop the math course, please do it before mid-term.
14. Take a walk keep your brain comfortable in mathematics
15. Good mathematician will thinking math at Pavilion of Harmony.
16. We strongly recommend you to make a beautiful, powerful note about all undergraduate math definition and theorems as well as proof. And then contribute to the society by open source and free for everyone. YOU WILL BE THE MATH JESUS!
17. Take a Guess.
18. LOSS is FUN.



0.3 Logic and Symbol

0.3.1 Story

As Alice was about to return to the real world, she decided to do something very brave...

She looked directly at the Mad Hatter and said: "Do you know that a crow is a writing desk?"

The Mad Hatter replied: "I don't know."

Alice was silent for a moment, then gathered her courage and said: "I like you."

The Mad Hatter was stunned for a moment and responded: "What does that have to do with anything?"

Alice stepped forward and gently kissed the side of the Mad Hatter's face: "So I like you."

After saying that, she immediately jumped into the rabbit hole, leaving the bewildered Mad Hatter behind.

--CTP inspired from <Alice in Wonderland>



0.4 Set Theory and Relation,Function

0.4.1 Story

In realms where Johnson and Huizi roam,
Above the sea of numbers, far from home,
Johnson spoke, "One plus one makes two,"
To which Huizi replied, "How can this be true?"

Thou art not two, how canst thou declare,
That one and one doth equal two with care?"
Johnson mused, "Thou art not I, 'tis clear,
How canst thou know what I hold dear?"

Huizi then countered, "I am not thee,
Thus, I cannot know thy certainty;
Thou art not two, so thy claim is frail,
What proof have we that two doth not prevail?"

Johnson urged, "Let us return to roots,
What dost thou know of harmony in truths?
If I know I know, then knowledge is mine,
Yet where is the sea that doth so align?"

Huizi asked, "What dost thou know of one?"
Johnson danced, avoiding the question spun.
In riddles they played, in wisdom's delight,
Two minds entwined in the depths of the night

0.4.2 ZFC

Axiom 1. Axiom of Weak Existence

There is a set.
 $(\exists x, x = x)$

Axiom 2. Axiom of Extensionality

Two sets are equal if they have the same elements.
 $(\forall x \forall y [\forall z (z \in x \iff z \in y) \implies x = y])$

Axiom 3. Axiom of Specification

Let $P(X)$ be a well formed formula with only one free variable X (set). For every set Z there exists a set Y such that X belongs to Y if and only if X belongs to Z and $P(X)$.
 $(\forall Z \exists S \forall x (x \in S \iff (x \in Z \wedge P(x))))$

Axiom 4. Axiom of Infinity

Given a set X , we define the successor $S(x)$ by $S(x) = x \cup \{x\}$

Axiom 5. Axiom of Choice

we have infinite bags, in each bags we can choose one sugar. There exist a function such that we can pick union of bags.
(Power set or union)

Axiom 6. Axiom of Pair

Given any sets x, y , there exist a set z which contains x and y elements.

$$\forall x \forall y \exists z [x \in z \wedge y \in z]$$

Axiom 7. Axiom of Union

Given any set X , there is a set Y such that for any element Z , Z is a element of Y if and only if there is a set W such that Z is element of W and W is element of X .

$$\forall x \exists y \forall z \forall w [z \in y \iff \exists w (z \in w \wedge w \in x)]$$

Axiom 8. Axiom of Power Set

Given any set x , there exists a set y such that given any set w , w is a subset of x if and only if every member of w is also an element of y .

$$\forall x \exists y [\forall w (w \subset x \implies w \in y)]$$

Axiom 9. Axiom of Choice

For any set X of nonempty sets, there exists a choice function f such that it is defined on X and maps each set of X to an element on that set.

"Every box contain at least 1 ball ,we can pick 1 ball in each box by Axiom of Choice"

$$\forall x [0 \notin x \implies \exists f : x \implies \bigcup x | \forall a \in x (f(x) \in a)]$$

Axiom 10. Axiom of Regularity(optional)

Every non-empty set A contains an element that is disjoint from A .

$$\forall x (x \neq \emptyset \implies \exists y (y \in x \wedge y \cap x = \emptyset))$$



0.4.3 Relations, Function and Ordering

Definition 1. Ordered Pairs

Let x, y be sets, the ordered pair of x and y is defined as the pair containing x and pair of both x, y .
 $(x, y) = \{x, \{x, y\}\}$.

Definition 2. Binary Relation

Let R be a binary relation on a set A where xRy refers to an ordered pair $(x, y) \in R$ for any $x, y \in A$.
 $\forall x \forall y (x, y \in A \implies (xRy \iff (x, y) \in R))$

Definition 3. Inverse Binary Relation

Let R be binary relation. Inverse of R is the set with element (y, x) for $(x, y) \in R$.
 $R^{-1} = \{z \mid z = (y, x) \text{ for some } x \text{ and } y \text{ such that } (x, y) \in R\}$

Definition 4. Domain

Domain of R is the set of all x which are in relation with some y .
 $domR = \{x \mid \text{there exists } y \text{ such that } xRy\}$

Definition 5. Range

Range of R is the set of all y which are in relation with some x .
 $ranR = \{y \mid \text{there exists } x \text{ such that } xRy\}$

Definition 6. Field

The set of domain R or range R is called the field of R .
 $domR \cup ranR = fieldR$

Definition 7. Image

The image of set A under R is the set of all y from the range of R related in R to some element of A .
 $R[A] = \{y \in ranR \mid \text{there exists } x \in A \text{ for which } xRy\}$

Definition 8. Inverse Image

The inverse image of B under R is the set of all x from the domain of R related in R to some element of B .
 $R^{-1}[B] = \{x \in domR \mid \text{there exists } y \in B \text{ for which } xRy\}$

Definition 9. Composition

Let R and S be binary relation. The composition of R and S is the relation below.
 $S \circ R = \{(x, z) \mid \text{there exists } y \text{ for which } (x, y) \in R \text{ and } (y, z) \in S\}$



Definition 10. Cartesian Product

Let x, y be two sets. there exist a set $x \times y$ whose elements are the ordered pairs (a, b) such that $a \in x$ and $b \in y$. (PLEASE PROVE IT)

Definition 11. Irreflexivity

Let R be a binary relation on a set A where xRy refers to $(x, y) \in R$ for any $x, y \in A$. R is irreflexive if $\forall x(x \in A \implies \neg xRx)$

Definition 12. Reflexive

For any set x and binary relation R , R is reflexive if x and x are in relation R .

$\forall x(x \in A \implies xRx)$

Definition 13. Irreflexivity

Let R be a binary relation on a set A where xRy refers to $(x, y) \in R$ for any $x, y \in A$. R is irreflexive if $\forall x(x \in A \implies \neg xRx)$

Definition 14. Transitive

For any set x, y and z . A binary relation R is transitive if x and y are in relation R , also y and z are in relation R then x and z are in relation R .

$\forall x \forall y \forall z (x, y, z \in A \implies [xRy \wedge yRz \implies xRz])$

Definition 15. Symmetric

For any sets x, y and binary relation R , R is symmetric if xRy then yRx .

$\forall x \forall y (x, y \in A \implies [xRy \implies yRx])$

Definition 16. Antisymmetric

For any sets x, y . A binary relation R is antisymmetric if x and y are in relation R , also y and x are in relation R , then $x = y$.

$\forall x \forall y (x, y \in A [xRy \wedge yRx \implies x = y])$

Definition 17. Asymmetric

For any $x, y \in A$. A binary relation R on A is asymmetric if xRy and yRx does not exist.

$\forall x \forall y (x, y \in A \implies [\neg (xRy \wedge yRx)])$

Definition 18. Connected

Let R be a binary relation and $x, y \in A$. R is connected if it is either xRy or yRx or $x = y$.

$\forall x \forall y (x, y \in A \implies (xRy \vee yRx \vee x = y))$

Definition 19. Comparable/Strong Connected

Let A be a set with a partial ordering R , and let $x, y \in A$. We say x and y are comparable and R is strong connected if xRy or yRx .

$\forall x \forall y (x, y \in A \implies xRy \vee yRx)$

Definition 20. Equivalence Relation

A binary relation E on a set A is an equivalence relation if it is reflexive, symmetric and transitive.

Definition 21. Ordering/(non-strict)partial order

Let A be a set, a binary relation R in A is a partial ordering (ordering) if it is antisymmetric, transitive and reflexive.

Definition 22. Poset

Let A be a set, A is partially ordering by R then we say ordered pair (A, R) as a poset.



Definition 23. Strict ordering/Strict partial Ordering

A binary operation R on a set A is a strict ordering if R is asymmetric, irreflexive and transitive.

Definition 24. Total Ordering/Linear Ordering

Let A be a set, and R be a partial ordering or strict ordering in A . R is a total ordering if R is strongly connected.

Definition 25. Chain

Let $B \subseteq A$ where A is ordered by $\leq$ where $\leq$ is a linear ordering. B is a chain in A if any two elements of B are comparable.

Definition 26. Well-Ordering

Let A be a set, and R be a partial ordering or strict ordering in A .

We say R is a well-ordering if

for any non-empty subset $B \subset A$, contains a least element with respect to R .

Definition 27. Least Element

Let $\leq$ be an ordering of A , and let $B \subseteq A$.

$b \in B$ is the least element of B in the ordering $\leq$ if $b \leq x$ for every $x \in B$.

$\forall A, \exists B \subseteq A [\min B = (b | x \leq b, \forall x \in B)]$

Definition 28. Minimal Element

$b \in B$ is the minimal element of B in the ordering $\leq$ if there exists no $x \in B$ such that $x \leq b$ and $x \neq b$.

Definition 29. Greatest Element

$b \in B$ is the greatest element of B in the ordering $\leq$ if $x \leq b$ for every $x \in B$.

Definition 30. Maximal Element

$b \in B$ is the maximal element of B in the ordering $\leq$ if there exists no $x \in B$ such that $b \leq x$ and $x \neq b$.

Definition 31. Lower Bound

$b \in B$ is the lower bound of B in the order set $(A, \leq)$ if $b \leq x$ for every $x \in B$.

Definition 32. Upper Bound

$b \in B$ is the upper bound of B in the order set $(A, \leq)$ if $x \leq b$ for every $x \in B$.

Definition 33. Partitions

A partition of a set A is a set S such that

1. Each $x \in S$ is a subset of A .
2. Any two sets $x, y \in S$ are disjoint or equal.
3. Any $z \in A$ is contained in some $x \in S$.

Definition 34. "Isomorphic of ordered sets"

An isomorphism between two ordered sets $(P, <)$ and $(Q, <)$ is a injective function h with domain P and range Q such that

$\forall p_1, p_2 \in P$

$p_1 < p_2 \iff f(p_1) < h(p_2)$

we also call $(P, <)$ and $(Q, <)$ are isomorphic.



Definition 35. Function

Let f be a function defined as $f : X \rightarrow Y$ where X is the domain of f and Y is the range of f . For any function f satisfy the first forbidden and second forbidden.

Definition 36. First Forbidden of Function

Let f be a function defined as $f : X \rightarrow Y$. For any $x \in X$ there exists a corresponding value of $y \in Y$ such that $f(x) = y$.

Definition 37. Second forbidden of Function

Let f be a function defined as $f : X \rightarrow Y$. For any $x \in X$, if x then there exist only one unique corresponding $f(x) = y \in Y$

Definition 38. Injective

f is injective if $f(x) = f(y)$ then $x = y$.

Definition 39. Surjective

f is surjective if for any $y \in Y$, then there exist $x \in X$ such that $f(x) = y$.

Definition 40. Bijective

f is bijective then f is injective and surjective.

Definition 41. "The Set of All Function Onto"

Let A and B be sets. The set of all function on A into B is denoted B^A .
(Prove it Exist)

Definition 42. Hasse Diagram

A Hasse Diagram is a graphical representation of the elements of ordering ($\leq$) or ($<$) with an upward orientation. A vertex (point) is drawn for each element of ordered set and join with line segment with the following rules ($\leq$ can be replaced by $<$):

- (a) If $p \leq q$ in the ordering then the point corresponding to p appears lower in the drawing than the point corresponding to q .
- (b) The two points of p and q will be joined by line segment if p is related to q .



0.4.4 Natural Number

Definition 43. Successor

The successor of a set x is the set $S(x) = x \cup \{x\}$

Definition 44. Axiom of Infinity

Given any sets x , we define the successor by $S(x) = x \cup \{x\}$
 $\exists x(\emptyset \in x \wedge \forall w[w \in x \implies S(w) \in x])$

Definition 45. Inductive set

A set I is called inductive if

- (a) $\emptyset \in I$
- (b) If $x \in I$ then $S(x) \in I$

Definition 46. Natural Number Set

The set of all natural number is the set contained in all inductive sets.
 $\mathbb{N} = \{x | x \in I \text{ for every inductive set } I\}$

Definition 47. Principle of Induction

Suppose $P(x)$ be a formula such that

- (a) $P(\emptyset)$ is true
 - (b) $P(x) \implies P(S(x))$ is true
- Then $P(x)$ holds for every $x \in \mathbb{N}$

Definition 48. Well-Ordering

A binary relation of R on a set A is well ordered if every non empty subset $B \subset A$ contains a least element and R is comparable(total) and transitive. Also R can be antisymmetric or asymmetric but not both.

Definition 49. Sequence

A sequence is a function whose domain is either $\mathbb{N}$ or a natural number n .

Definition 50. Parametric Recursion

For any set A , any $a \in A$, and any function $g : A \times \mathbb{N} \rightarrow A$, there exists a unique sequence $f : \mathbb{N} \rightarrow A$ such that

- (a) $f_0 = a$
- (b) $f_{n+1} = g(f_n, n)$ for all $n \in \mathbb{N}$

(PROVE IT)



0.4.5 Ordinal Number

Definition 51. *Initial Segment*

Let $(W, <)$ be a well ordered set, and let $a \in W$. $W[a]$ is an initial segment if $W[a] = \{x \in W \mid x < a\}$

Definition 52. *(Strict) Well Ordering of $\in$*

Let $(L, <)$ be a linearly-ordered set. An initial segment of $(L, <)$ is a subset $S \subset L$ such that $S \neq L$ (S is a proper subset) and $a \in S$ implies that there exists a $x \in S$ for all $x < a$.

Definition 53. *Transitive with $\in$*

Let $\in$ be a binary relation.

A set A is transitive if $x \in A$ and $y \in x$ implies that $y \in A$.

$\forall A (x \in A \text{ and } y \in x \implies y \in A)$

Definition 54. *Ordinal*

A set α is an ordinal number if

1. α is transitive

2. α is well ordered by $\in$

We say ordinal instead of ordinal number too.

Definition 55. *Successor Ordinal*

A successor ordinal is an ordinal x such that $x = S(y)$ for some ordinal. A limit ordinal which is not a successor.



0.4.6 Theorem about Axiomatic Set Theory

Theorem 1. Russell Paradox

Proof. Let R be a set, and we define it as $R = \{x | x \notin x\}$. Suppose $R \in R$ then $R \notin R$. Suppose $R \notin R$ then $R \in R$. It is a tautology. ■

Theorem 2. Principal of Explosion

Proof. For any statement P and Q , if P and $\neg P$ is true then it follows that the two part statement Q is true, the universe will EXPLODE!!!! Therefore P and $\neg P$ cannot be true.

Suppose (P) there is a universe and $(\neg P)$ there is not a universe, both statements are assumed to be true. Then Q second part statement is that "universe will EXPLODE! or there is a universe". Since we know that P state out there is a universe then Q is true since Q is a union statement. However $\neg P$ state out that P is not true, there is no universe. Therefore UNIVERSE WILL BE EXPLOSIONSSSSSSSS by the remaining part of Q . ■

Theorem 3. Existence of Empty Set (Existence Axiom)

Proof. By the axiom of weak existence, there exists a set A . Then by the axiom of specification, there is a set E such that $\forall x(x \in E \iff x \in A \wedge x \neq x)$ where the property is $P(x) = \{x | x \neq x\}$. Choose any x since it is a tautology that $x \neq x$, we have $x \in E$. Hence we show the existence of empty set. ■

Theorem 4. Existence of Union Set

Proof. Let X be a set. By axiom of power set, a set Z such that given any set z , z is an element of u iff there is a set w such that z is element of w and w is a element of x . By Axiom of Specification, define $U = \{x \in Z | x \in y \text{ for some } y \in x\}$. Then U is an union. ■

Theorem 5. Existence of Intersection Set

Proof. Let x and y be a set.

Recall that $\cup x = \{z | \exists y(z \in y \wedge y \in x)\}$ we define the intersection as

$\cap x = \{z \in \cup x | \forall y(y \in x \rightarrow z \in y)\}$ by axiom of specification and axiom of union. ■

Theorem 6. Existence of Power Set

Proof. Let Z, x be a set and define $w \subset x$. There exist a set $P(x) = \{w | x \in Z \wedge w \subset x\}$ by axiom of Specification and axiom of power set. ■

Theorem 7. Existence of the Complement Set

Proof. Let X and Y be a set. Suppose $X \subseteq Y$ Then $Y \setminus X$ is called the complement if $Y \setminus X := \{y \in Y | x \notin X\}$ by axiom of specification. ■



Theorem 8. *Set of Product and Finite Dimension Cartesian Product*

Proof. Let x, y be sets. Show that the set $x \times y$ exists. Suppose $x \in X$ and $y \in Y$. Then we define $X \times Y = \{(x, y) \in P(P(X \cup Y)) \mid x \in X \text{ and } y \in Y\}$. Since $\{\{x\}, \{x, y\}\} = (x, y)$ by definition of ordered pair, then $\{\{x\}, \{x, y\}\} \in P(P(X) \cup P(X \cup Y)) = P(P(X \cup Y))$. Therefore $X \times Y$ exists.

Then we can define the set $X \times Y \times Z$ exists, we define

$X \times Y \times Z = \{(x, y, z) \in P(P(X \cup Y \cup Z)) \mid x \in X \text{ and } y \in Y \text{ and } z \in Z\}$. Note that $\{\{x\}\{x, y\}\{x, y, z\}\} \in P(P(X) \cup P(X \cup Y) \cup P(X \cup Y \cup Z))$ and other power set are all subset of $P(X \cup Y \cup Z)$. Therefore we define the Triple Cartesian Product.

We will show the set $\prod_{u \in z} u$ exists by contradiction. We define u^z as the set of function with domain z with target u . Then $f \in u^z$ such that $\forall u \in z, f(u) \in u$ by axiom of comprehension. When z is finite, it coincide with our naive idea of the product. ■

Theorem 9. *D.N.E the Set of All Set (Regularity Axiom)*

Proof. Suppose there exist a set A that contains all sets. However if we define the set of Russell Paradox then $R \notin A$. Contradiction arises, there does not exist the set of all set. ■

Theorem 10. *Sets, X, Y, Z Exists Pair Contains X and Y but not Z* **Theorem 11.** *Existence of Union X and Y but not contains Z* **Theorem 12.** *Use Partition determines an equivalence relation***Theorem 13.** *Proof the Composition of Function*

Proof. (CLAIM): We claim that the composition $g \circ f : A \rightarrow D$ is well defined.

Let $f : A \rightarrow B$ and $g : B \rightarrow C$.

Since $f(x)$ is the element of B for any $x \in A$ then $g(f(x))$ is well-defined by the domain of g is also B . Note that $g(f(x))$ is an element of C too, then it must be unique on C for any $x \in A$ by first forbidden of function. Therefore $g \circ f : A \rightarrow C$ is well-defined. ■

Theorem 14. *Existence of Infinite Composition of Composition***Theorem 15.** *Existence and Uniqueness of least element***Theorem 16.** *Ordering and natural number " $<$ " " $>$ "*

Theorem 17. $\mathbb{N}$ is an inductive set and $\mathbb{N}$ is smallest inductive set

Proof. Since $\mathbb{N}$ is the subset of any inductive set as any set is the subset of itself. We want to prove that $\mathbb{N}$ is the smallest inductive set which means $\mathbb{N} \subset I$ for any inductive set I .

Let I', I be an inductive sets. Suppose $I \cap I' \neq \emptyset$. Then $I \cap I'$ is also inductive set since $0 \in I \cap I'$ and $\forall x \in I \cap I' \implies S(x) \in I \cap I'$ by the inductive property of I, I' . Since $\mathbb{N}$ is the subset of all inductive set then $\mathbb{N} \subset I \cap I' \subset I$. Also $\mathbb{N} \subset I \cap I' \subset I'$. Therefore N is the subset of any arbitrary inductive set. It implies that N is the smallest inductive set. ■

Theorem 18. Natural Number smaller then $n+1$ then it equals to n or smaller then n

Theorem 19. Natural Number equal to some natural number add another natural number

(STATEMENT:) $\forall n, m \in \mathbb{N}$, If $m < n$ then there exists some unique $k \in \mathbb{N}$ such that $m = n + k$.

(PROOF:)

Theorem 20. $(\mathbb{N}, <)$ is linearly ordered

Proof. We want to show that $(\mathbb{N}, <)$ is comparable, asymmetric, transitive as it is a strict linearly ordered by $<$ intuition.

We will use the Principal of Induction to prove the transitive of $\mathbb{N}$.

Let $P(n)$ be the statement that

if $k < m$ and $m < n \implies k < n$ for any $k, m, n \in \mathbb{N}$.

We first show the base case.

Let $n = 0$ then $k < m$ and $m < 0$ which is impossible as $k, m \in \mathbb{N}$ and 0 is the least element of N . Since the condition false then the statement become true by logic. Therefore $P(0)$ is true.

Suppose $P(l)$ is true. Then $k < m$ and $m < l$ implies $k < l$.

We would love to show $P(l+1)$ is true.

Suppose $k < m$ and $m < l+1$. Then by assumption part $k < m < l$ so now we have $m < l$ and $m = l$ by Lemma last part.

If $m < l$ then $k < l$ by induction. If $m = l$ then $k < l = m$ by conditional part. Hence $P(n)$ is true and $\mathbb{N}$ is transitive.

Now we will prove that $(N, <)$ is asymmetric by Principal of Induction.

Suppose $n < k$ and $k < n$ and this implies that $n < n$. Then we want to show that it is impossible by induction. For $n = 0$, we have $0 < 0$ is impossible. Let us assume that $n < n$ is false then for $n+1$ case, we will prove by contradiction.

Suppose $(n+1) < (n+1)$ then it is either $n+1 < n$ or $n+1 = n$. Since $n < n+1$ is either $n = n$ or $n < n$. Note that $n < n$ is false and we only use relation $<$. Therefore $n+1 < n+1$ cannot hold.

Hence $P(n)$ is true. Thus we can conclude that $\mathbb{N}$ is asymmetric.

Now we finally prove that $\mathbb{N}$ is strongly connected. In other words, $m < n$ or $n < m$ or $m = n$ for any $m, n \in \mathbb{N}$. We will prove it by induction. For $n = 0$, we have $m < 0$ or $0 < m$ or $m = 0$ for any $m \in \mathbb{N}$, first case impossible, but second case and third case is possible. Then $P(0)$ holds. Suppose $P(k)$ is true, then $m < k$ or $k < m$ or $k = m$ for any $m \in \mathbb{N}$.

For $P(k+1)$, If $m < n$ in assumption part then $m < n+1$ follows immediately by transitivity of $<$. If $m = n$ by assumption part then $m < n+1$ by definition $<$. If $n < m$ then we would like to conclude that $(n+1) \leq m$. We need to show that it is either $n < m$ or $m = n$ as The assumption part leads immediately $n < m$ is true. Therefore $P(n)$ is true. Thus $\mathbb{N}$ is strongly connected.

Hence $(\mathbb{N}, <)$ is total ordering and thank you for your patient to go through this magical journey with us. ■



Theorem 21. *Inductive Theorem Second Version*

Theorem 22. $\mathbb{N}$ is well ordered

Proof. We already know that $\mathbb{N}$ is total ordering, now we just need to prove that every nonempty subset of $\mathbb{N}$ contains least element. We will prove it by Contradiction.

Let X be a non-empty subset of $\mathbb{N}$. Suppose X does not contain the least element. Then let us consider $\mathbb{N} \setminus X$ contains least element. PROVE BY MAGIC, we observe that if $k \in \mathbb{N} \setminus X$ for all $k < n$ where n is the largest element then $n \in \mathbb{N} \setminus X$, otherwise n would be the least element of X . BY THE SECOND INDUCION Principle, we conclude that $n \in \mathbb{N} \setminus X$ holds for any natural number n . This leads to $X = \emptyset$ and X is non-empty set. Contradiction arises! Hence $\mathbb{N}$ is well-ordered set. ■

Theorem 23. *Prove $\mathbb{N}$ Addition*

Proof. Let f be a function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ such that

(a) $+(m, 0) = m$ for any $m \in \mathbb{N}$

(b) $+(m, n+1) = +(m, n) + 1$ for any $m, n \in \mathbb{N}$

The function is well-defined by Parametric Reduction.

Let $A = P = \mathbb{N}$ and $a(p) = p$ for all $p \in \mathbb{N}$ and $g(x, p, n) = x + 1$ for all $x, p, n \in \mathbb{N}$. First we will check condition (a)

Pick $n = 0$ then $+(m, 0+1) = +(m, 0) + 1$ and $+(m, 0) = m$. Also we have $0 + 1 = S(0) = 1$ by definition of number 1. Therefore we have $+(m, 1) = m + 1 = S(m)$ and the successor of $m \in \mathbb{N}$ is actually the sum of m and 1. Now we call this function as addition and we prove the existence of addition of $\mathbb{N}$ and write $+(m, n) = m + n$

We will prove that $m + (n + 1) = (m + n) + 1$ by Principal of Induction.

Let $P(l)$ be the proposition that "For any $n, m \in \mathbb{N}$, $(n + m) + l = n + (m + l)$ "

Then by properties of 0,

$(n + m) + 0 = n + m = n + (m + 0)$ so that $P(0)$ is true

Suppose $P(k)$ is true. Then $(n + m) + r = n + (m + r)$.

Now we will prove $P(r + 1)$

$(n + m) + r + 1 = ((n + m) + r) + 1 = (n + (m + r)) + 1 = n + ((m + r) + 1) = n + (m + (r + 1))$

Hence $P(r + 1)$ is true.

Prove $m + 0 = m$ by induction. Let $P(n)$ be the proposition that "For any $n \in \mathbb{N}$, $n + 0 = n$." For $n = 0$ then $0 + 0 = 0$ by (a) condition.

Suppose $n = k$ is true then $k + 0 = k$

For $n = 1 + k$, $(1 + k) + 0 = 1 + (k + 0) = 1 + k$ by the last statement and associativity. Therefore we prove the commutative law of addition. (REMAINLY TO PROVE ASSOCIATIVITY) ■



Theorem 24. *Prove $\mathbb{N}$ Subtraction*

Proof. We define the subtraction $- : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ as follows

(a) $n - 0 = n$ for any $n \in \mathbb{N}$

(b) $m - (n + 1) = m - n - 1$ for any $m, n \in \mathbb{N}$ Subtraction function is well defined by Parametric Recursion. ■

Theorem 25. *Prove $\mathbb{N}$ Multiplication*

Proof. We define the multiplication $\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ as follows

(a) $n \cdot 0 = 0$ for any $n \in \mathbb{N}$

(b) $m \cdot (n + 1) = m \cdot n + m$ for any $n, m \in \mathbb{N}$

Then by Parametric Recursion we define this function is well-defined and we call it as multiplication. ■

Theorem 26. *$<$ is a strict ordering, there is one and only one partial ordering $\leq$ such that $a < b$ if and only if $a \leq b$ and $a \neq b$*

Theorem 27. *If $\leq$ is a partial ordering then there is one and only one total ordering such that $a < b$ if and only if $a < b$ or $a = b$*

Theorem 28. *For any $x, y, z \in \mathbb{N}$. If $x \in y$ and $y \in z$ then $x \in z$*

Theorem 29. *Non-strict Well Ordering is non strictly total ordering*

Proof. Let A be a set and R be a (non strict) partial ordering on A .

(CLAIM:) Suppose A is well ordered by R . Then A is a total ordering.

(PROOF:) Let A be a set and R be a (non strict) partial ordering on A .

Suppose A is well ordered by R . Then for any non empty subset $B \subset A$ has a least element with respect with R by definition of Well-Ordering.

We can restrict R as $\leq$ by the least element property.

We want to show that R is strongly connected ($\forall a, b \in A, aRb$ or bRa).

Pick any $a, b \in A$. We define $\{a, b\}$ as a set by the axiom of specification and axiom of pair as well as axiom of weak existence. Note that $\{a, b\}$ is a subset of A . Then $\{a, b\}$ has a least element by well ordered A . We can pick a or b as the least element.

By the definition of least element, if a is the least element then $a \leq b$, also if b is the least element then $b \leq a$. Therefore $a \leq b$ or $b \leq a$ in set $\{a, b\}$. Note that A is strongly connected and since $R = \leq$ is a partial ordering. Then A is a total ordering.

(SAME PROOF AS STRICT WELL ORDERING AND STRICT TOTAL ORDERING, Left as exercise) ■



Theorem 30. *Ordinal cannot contains itself*

Proof. Note that $\in$ defines as a strict ordering so it is irreflexivity $\implies a \notin a$ ■

Theorem 31. *If something belongs to ordinal then it is ordinal too*

Proof. Let α be a ordinal and $x \in \alpha$

We will first check the transitivity of x .

Let $y \in x$ and $z \in y$. Then $z \in y \in x$ such that $z \in x$. Therefore x is transitive.

We will verify that x is well ordered.

For any non empty subset y of x , $y \subset x \subset \alpha$ since α is well ordered so y has a least element too.

Therefore x is ordinal. ■

Theorem 32. *Successor of Ordinal is ordinal too!*

Proof. (CLAIM:) If x is a ordinal then $S(x)$ is also a ordinal.

Let x be a set. Suppose x is a ordinal and $S(x) = x \cup \{x\}$.

We want to verify $S(x)$ is transitive and well ordered by $\in$.

We will first verify that $S(x)$ is transitive.

Pick any $\alpha \in S(x)$. Then $\alpha \in x$ or $\alpha \in \{x\}$ by the definition of successor.

Suppose $\alpha \in x$ and let $\beta \in \alpha$. Then $\beta \in x$ by the transitivity of ordinal x . Therefore $\beta \in x \cup \{x\} = S(x)$.

Suppose $\alpha \in \{x\}$ and let $\beta \in \alpha$. Then $\beta \in \{x\}$ by the transitivity of ordinal x . Therefore $\beta \in x \cup \{x\} = S(x)$.

We show that $S(x)$ is transitive.

Now we will show that $S(x)$ is strict well ordering by $\in$.

Let A be an non empty set and $A \subset S(x) = x \cup \{x\}$.

Suppose $A \subset x$. Then there exists a least element $\delta \in A$ such that $\alpha < \delta$ for any $\alpha \in A$ by well ordering of x by $\in$.

Suppose $A \subset \{x\}$. Then $A = \{x\}$ or $A \subsetneq \{x\}$.

If $A = x$ then obviously the least and only one element is x such that A is well ordered.

If $A \subsetneq \{x\}$ then $A \setminus \{x\} \neq \emptyset$. There is a least element say $\delta \in A \setminus \{x\}$ since $A \setminus \{x\} \subset x$ and x is well-ordered.

Hence $S(x)$ is well-ordered by $\in$. ■

Theorem 33. *Every Natural Number is Ordinal*

We will prove it by Principle of Induction.

Suppose $P(n)$ be the statement such that n is ordinal number for every $n \in \mathbb{N}$

Considering Base Case as the empty set, $\emptyset$ is transitive since no such set $x \in \emptyset$.

$\emptyset$ is well ordered by $\in$ since there is no non-empty set of $\emptyset$.

Suppose $P(k)$ is true for some $k \in \mathbb{N}$, then k is transitive and well ordered by $\in$.

We will now show that $P(S(k))$ is transitive too.

Since $k \in \mathbb{N}$ and for any $y \in k$, $y \in \mathbb{N}$ by the assumption case of $P(k)$.

Let $a \in S(k) = k \cup \{k\}$ then $a \in k$ or $a = k$. Therefore $a \in \mathbb{N}$.

Hence $\mathbb{N}$ is transitive.

Now we will verify that $\mathbb{N}$ is strict well ordered by $\in$.

Let non empty set $A \in \mathbb{N}$ and define $A' = \mathbb{N} \setminus A$.

Now we will try to Prove by Contradiction.

Suppose A does not have a least element. Then $\emptyset \in A'$. Otherwise $\emptyset \in A$.

Since $A' \in \mathbb{N}$ then $\emptyset, S(\emptyset), S(S(\emptyset)), \dots, n, S(n) \in A'$ by the definition of $\mathbb{N}$.

We observe that A' is a inductive set as empty set $\in A'$ and successor of $n \in A'$.

A' is the set of all inductive set such that $\mathbb{N} \in A'$ such that $\mathbb{N} = A$.

This leads to $A = \emptyset$. Contradiction arises, Therefore A contains least element.

Hence $\mathbb{N}$ is an ordinal.



0.4.7 Dialogue

Dialogue 1. We want to use the most SIMPLE axiomatic version of ZFC so that we won't include AXIOM OF REGULARITY and AXIOM OF EXISTENCE as we can prove those "axioms".

Dialogue 2. Every elements and symbols (x,y,z) are all sets so don't confuse set or elements or member or number. We preserve everything as sets as we are SET THEORY.

Dialogue 3. MC BEAN Where is the ISOMORPHIC. Left as EXERCISE 29. "Guess the definition of an "isomorphism of ordered sets"????"

Dialogue 4. You can forget the details of axiom of choice as it is hard to test in MID TERM and I didn't defined what is function so the axiom is not rigorous enough.

Dialogue 5. With the help of Yuki ShiJie and Kelvin helps, we know that strict ordering is not reflexive as it contradicts the definition of asymmetric.

Dialogue 6. With the help of our new TA Kelvin, he points out that strict ordering is not ordering, what a INSPIRING point of view!

Dialogue 7. By our Chairman Kelvin Ho, He points out that initial segment must be j but not $\leq$ in definition as it seems to contradict the well ordered set definition.

Dialogue 8. One day CTP has a huge problem of Well Ordering with (N, j) as j is not antisymmetric $\implies j$ is not a ordering $\implies j$ is not a linear order $\implies j$ is not a Well Ordering in Textbook and Mc Bean notes. Fortunately, Kelvin and Google Chrome helps poor Marco. We figure out that Well Ordering only focus on least element and total(comparable) property so it satisfy 2 version of inequality($<$) and($\leq$) so we don't care reflexive or antisymmetric even asymmetric, so CTP improve the definition of Well Ordered.

Dialogue 9. CTP find out that set contains less then 2 element can still holds the transitive as if aRa is true then aRa and $aRa \implies aRa$.

Dialogue 10. CTP find out that antisymmetric and symmetric can holds together for some special case during the MINDFUCK of Well Ordered.

Consider binary relation " $=$ ".

$=$ is symmetric as $a = b \implies b = a$

$a = b$ and $b = a \implies a = b$

Therefore $=$ is both antisymmetric and symmetric

Dialogue 11. CTP is really dumb and he curious about the world of mathematics so he ask these questions (a) There exists relation both symmetric and asymmetric? If no prove it.

(b) There exists relation holds both anti-symmetric and asymmetric? If no prove

(c) There exists relation holds for symmetric, antisymmetric and asymmetric?

Dialogue 12. the author was inspired by Dear Dr Fong Note so I strongly recommend you to have a taste of Dr Fong rigorous prove style.

Dialogue 13. 13. Maybe one day we can develop a website or platform to let everyone in the world to get free, libre and Open Source notes about Modern Mathematics and I would like to draw a comics or games or animation to develop math memes? (Inspired by Chan Kai Leung)

Dialogue 14. Asymmetric implies irreflexivity.

Dialogue 15. By Yuki a strict total order j can be defined by 2 equivalent ways:

(a) $a < b$ if $a \leq b$ and $a \neq b$ (reflexive reduction)

(b) $a < b$ if not $b < a$ (i.e. j is the complement of the converse of $\leq$)

Dialogue 16. Comparable uses to describe two element in A but not the binary relation R .

Dialogue 17. Credit the proof of ordinal to Yuki (Cheung Yuk Ki).



Dialogue 18. *It seems like we never use axiom of regularity but we use this to claim there cannot exist set that contains itself and no infinity sequence a_{i+1} as a_i element*

Dialogue 19. *Ordinal's well ordering property has to be strict as it define well ordering on j . If we define on $\leq$, then we will contradict to the axiom of regularity and makes something like $3 \in 3$.*

Dialogue 20. *Credit the composition proof of function to JoJo and Yuki. CTP mainly make the proof more clearly as it is quite frustrated for dumb CTP.*

Dialogue 21. *Credit the GREAT PROOF about $\mathbb{N}$ is inductive set from JoJo and textbook.(P.S. she is chem major but I think she is better than half of math major as she read textbook)*

Dialogue 22. *Yuki Comment:*

Function on A means its domain is A .

Function into B means that there exists some element, say a , belongs to A such that a maps to its range but not its codomain, like B (must not be a surjective function from a domain to B).

Function onto B means B is the range of the function F ,

which means F is surjective. In other words for any $b \in \text{range } B$, there exists a x from its domain such that $f(a) = b$.

(WHAT A INSPIRING MOMENT, I SHOULD RETAKE MATH 1050)

Dialogue 23. *Jojo comment : So a function is not an onto function iff it is an into function.*



0.5 Natural Number, Real Number, Complex Number



0.6 Algebraic Structure



0.7 Linear Algebra



0.8 Abstract Algebra



0.9 Discrete Mathematics



0.10 Field and Galois Theory



0.11 Modules and Representation Theory



0.12 Number Theory



0.13 Graph Theory



0.14 Topology



0.15 Basic Analysis



0.16 Calculus



0.17 Advanced Calculus

0.18 Real Analysis



0.19 Complex Number and its application



0.20 Complex Analysis



0.21 Functional Analysis



0.22 ODE and PDE



0.23 Numerical Analysis



0.24 Fourier Analysis



0.25 Probability Theory



0.26 Statistics



0.27 Stochastic Progressing



0.28 Mathematics Modeling



0.29 Data Analysis

0.29.1 Statistics for Data Analysis

Definition 56. *Bayes Rule*

$$P(A|B) = \frac{P(A)}{P(B)} P(B|A)$$

Definition 57. *Gaussian Distribution/Normal Distribution*



0.29.2 Linear Algebra for Data Analysis

Definition 58. *Linear Independence*

Definition 59. *Dimension*

Definition 60. *Basis*

Definition 61. *Rank*

Definition 62. *Null Space*

Definition 63. *Eigenvalues and eigenvectors*

Definition 64. *Rotation and Projection*

Theorem 34. *Singular Value Decomposition (S.V.D)*

Every $m \times n$ matrix A can be decomposed as $U \Sigma V^*$ where Σ is a diagonal matrix and U, V are unitary whose's columns are orthonormal ($U^H = U^{-1}$) where U^H is conjugate transpose. Note that A^* is the adjoint of A

(a) Check the rank of matrix A

(b) Find the λ eigenvalues of A

(c) Find the singular values or eigenvector v_i of A^*A and AA^* , all equal to $\sigma = \sqrt{\lambda}$

(d) Find the corresponding orthonormal eigenvector to A which is same as the eigenvector of A^*A for different eigenvalues. Also normalize the eigenvectors.

(e) $\Sigma = \text{diag}\{\sigma_i\}$ for any i corresponding to the eigenvalues, also we will put the larger one in the upper row. (f) V^* is the matrix

of corresponding eigenvector also remember $V^* = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

(g) Then find out the U through calculation.

Example 1. Find the S.V.D. of matrix $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$ Note that the rank of A is two as there are 2 linearly independent row in matrix A . Then there are 2 eigenvalues and we determine signal values σ_1 as the bigger one corresponding to λ_1 . We will see that σ_1 lagger then λ_1 and σ_2 less then λ_2 . Since $\lambda_1 = 3$ and $\lambda_2 = 5$, let us find the σ_1 and σ_2 :

$A^T A = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$ or $AA^T = \begin{pmatrix} 9 & 12 \\ 12 & 41 \end{pmatrix}$ their characteristic equitation are same as $(\lambda - 45)(\lambda - 5)$

so that $\sigma_1 = \sqrt{\lambda_1} = \sqrt{45}$ and $\sigma_2 = \sqrt{\lambda_2} = \sqrt{5}$

Then we find out

$\begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 45 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ Since their length are 2 then we find out the normalized vector $v_1 = \frac{1}{\sqrt{2}}$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ Then we compute $Av_1 = \frac{3}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{\sqrt{45}}{\sqrt{10}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \sigma_1 u_1$ and $Av_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -3 \\ 1 \end{pmatrix} = \frac{\sqrt{5}}{\sqrt{10}} \begin{pmatrix} -3 \\ 1 \end{pmatrix} = \sigma_2 u_2$

Hence $U = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{pmatrix}$ as well as $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$



0.29.3 Regression.1

Definition 65. Regression

Regression is a statistical model to determine the relationship between a dependent variable and one or more independent variables.

Definition 66. Linear Regression

Linear Regression is a statistical model that estimate the linear relationship between dependent variable and one or more independent variables.

Definition 67. Data Set

Data set is a collection of data defined as a set $D := \{(x_i, y_i)\}_{i=1}^n$ where inputs $x_i \in \mathbb{R}^d$ and target variable $y_i \in \mathbb{R}$.

Definition 68. True Function

True Function is the true relationship between inputs x and output y , we define as $f(x) = y$ which is normally unknown. If f is known then we can sleep now as we don't need to do any regression analysis.

Definition 69. Hypothesis Function

Hypothesis Function aims to approximate true function $\hat{f}$, we define it as

Ideas 1. Linear Regression aims to find cost function $\mathbb{L}$ and best parameter w^* .

Given any data set $D := \{(x_i, y_i)\}_{i=1}^n$, for any input $x \in \mathbb{R}^d$ and output $y \in \mathbb{R}$. We want find the true function $f(x) = y$ using hypothesis function $h(x) = \tilde{p}$ to approximate $f(x) = y$. We define hypothesis class $\mathbb{H} = \{h_w | w \in \mathbb{R}^d\}$ of hypothesis functions. We also employ a parametric model such that there is some finite-dimensional vector $w \in \mathbb{R}^d$. The element w is a parameters or weights to control the hypothesis function and then approximate true function. Hence $h_w(x) = g(x, w)$.

Linear Regression aims to design a cost function ($\mathbb{L}$) to find the best parameter w^* such that $w^* = \arg_w \min \mathbb{L}(w)$.

Ideas 2. Ordinary Least Squares

We want to find the best parameter w^* to approximate the true function so that we use ordinary least square, it aims to minimize the sum of the squares of the residuals. In other words it aims to minimize the sum of the squares of the offsets of the point from the curve. To do so, we need to make sure X is full rank and given data X, y so if it is not full rank then the optimal w_{OLS}^* is not unique. Since X being full rank means that each columns of X is a one-to-one correspondence between w and Xw .



Theorem 35. *Ordinary Least Squares (O.L.S)*

If X is full rank then the optimal parameter of ordinary least squares is $W_{OLS}^* = (X^T X)^{-1} X^T y$

Proof. Ordinary Least Squares (O.L.S) is the simplest regression method, we take $h_w = x^T w$ such that

$$y_i \approx \tilde{y}_i = h_w(x_i) = x_i^T w \iff \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \tilde{y}_1 \\ \vdots \\ \tilde{y}_n \end{pmatrix} = \begin{pmatrix} x_1^T & \dots & x_n^T \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_d \end{pmatrix} \iff y = Xw$$

where the design matrix $X \in \mathbb{R}^{n \times d}$ contains datapoint x_i as its i th row. Usually $n \geq d$ meaning there are more datapoints than measurements. Therefore there is no exact unique solution to the equation $y = Xw$, but we can find an approximate solution by minimizing the sum of the squared errors: $L(w) = \sum_{i=1}^n (x_i^T w - y_i)^2 = \min_w \|Xw - y\|$.

Now we will find the particular structure of OLS allows us to compute a closed-form expression for a globally optimal solution, which we denote W_{OLS}^* .

We will use calculus as the first approach to find optimal parameter W_{OLS}^* .

Let cost function be $L : \mathbb{R}^d \rightarrow \mathbb{R}$ which is continuously differentiable,

then any local optimum w^* satisfies $\nabla L(w^*) = 0$, also note that $(Xw)^T = y^T$ and $Xw = y$ such that

$$\begin{aligned} L(w) &= \|Xw - y\|_2^2 = (Xw - y)^T (Xw - y) \\ &= (Xw)^T (Xw) - (Xw)^T y - y^T (Xw) + y^T y \\ &= w^T X^T X w - 2w^T X^T y + y^T y \end{aligned}$$

Using the following results from matrix calculus:

$$\nabla_x (a^T x) = a$$

$$\nabla_x (x^T A x) = (A + A^T)x$$

Now we will do the partial differentiation as follows

$$\begin{aligned} \nabla L(w) &= \nabla_w (w^T X^T X w - 2w^T X^T y + y^T y) \\ &= \nabla_w (w^T X^T X w) - 2\nabla_w (w^T X^T y) \\ &= 2X^T X w - 2X^T y \\ &= 0 \end{aligned}$$

Since X is full rank then $(X^T X)$ is invertible such that $W_{OLS}^* = (X^T X)^{-1} X^T y$.

Now we will use orthogonal projection as the second approach to find optimal parameter W_{OLS}^*

Let V be an inner product space and S be a subspace of V , then any $v \in V$ can be decomposed uniquely in the form $v = v_S + v_\perp$ where $v_S \in S$ and $v_\perp \in S^\perp$. Here $S^\perp$ is the orthogonal complement of S , so $S^\perp \perp S$, the vector of $S^\perp$ are all perpendicular to every vector in S .

The orthogonal projection onto S denoted P_S , is the linear operator that maps v to v_S in the decomposition above.

Note that $\|v - P_S v\| \geq \|v - s\|$ for all $s \in S$, with equality if and only if $s = P_S v$. That is $P_S v = \arg_{s \in S} \min \|v - s\|$

Now let us considering the case of O.L.S, $w_{OLS}^* = \arg_w \min \|Xw - y\|_2^2$

Note that the set of vector Xw for some $w \in \mathbb{R}^d$ is precisely the range of X , which is the subspace of $\mathbb{R}^n$ as $\mathbb{R}^d \subset \mathbb{R}^n$, so

$\min_{z \in \text{ran}(X)} \|z - y\|_2^2 = \min_{w \in \mathbb{R}^d} \|Xw - y\|_2^2$. Note that $P_{\text{ran}(X)} y = Xw_{OLS}^*$ where w^* is any optimum for the right-hand side.

The projected point Xw_{OLS}^* is always unique and w^* is unique too if X is full rank. We now want to solve for w_{OLS}^* , we need the following fact: $\text{null}(X^T) = \text{range}(X)^\perp$

Since we are projecting onto range X , the orthogonality condition for optimality is that

$y - Py \perp \text{range}(X)$ and $y - Xw \in \text{null}(X^T)$. Therefore $X^T(y - Xw_{OLS}^*) = 0$ such that $W_{OLS}^* = (X^T X)^{-1} X^T y$. ■



Remark 1. *The calculus approach is suck as we might not know the matrix calculus.*

Remark 2. $\nabla L(w) = 0$ only means us find a critical point, it maybe local maximum or local minimum, or a saddle point, so why we can confirm it is global minimum point???

Due to the convex cost function L . If we want to show its convex then we need to compute the Hessian of L which is the second partial order matrix. Note that $\nabla^2 L(w) = 2X^T X$

It shows that this is positive semi-definite as symmetric matrix $w^T (X^T X) w$ eigenvalues are all nonnegative :
 $w^T (2X^T X) w = 2(Xw)^T Xw = 2\|Xw\|_2^2 \geq 0$

Remark 3. Prove $P_s v = \arg_{s \in S} \min \|v - s\|$. Then by the Pythagorean Theorem,
 $\|v - s\|^2 = \|v - P_s v + P_s v - s\|^2 = \|v - P_s v\|^2 + \|P_s v - s\|^2 \geq \|v - P_s v\|^2$

Example 2. For any $x_1 = \begin{pmatrix} 17 \\ 13 \\ 1 \end{pmatrix}$ $x_2 = \begin{pmatrix} 24 \\ 16 \\ 1 \end{pmatrix}$ $x_3 = \begin{pmatrix} 27 \\ 0 \\ 1 \end{pmatrix}$ $y = \begin{pmatrix} 14393 \\ 13405 \\ 8316 \end{pmatrix}$ we have $X = \begin{pmatrix} x_1^T \\ x_2^T \\ x_3^T \end{pmatrix} = \begin{pmatrix} 17 & 13 & 1 \\ 24 & 16 & 1 \\ 27 & 0 & 1 \end{pmatrix}$

By O.L.S we find out $W_{OLS}^* = (X^T X)^{-1} X^T y = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$

Remark 4. *I'm too lazy to calculate it and I ask AI to do it 3 time still wrong as $Xw \neq y$.*

Code 1.

Ideas 3. *Disadvantage of O.L.S*

There are a lot of bugz on O.L.S, it only solve linear least square problem and it falls short due to numerical instability and generalization issues: (X not full rank, sensitive to outliers, colinear make its not stable) It will b overfitting and multicollinearity. Overfitting means the math model accurates to the dataset but bad at approximate things. Multicollinearity cases the matrix X losing its rank and full rank of X is really important to O.L.S. How about small singular values very close to 0? Very small singular value cases the squared inverse of the the singular values of X being extremely large, it will be a huge problem of numerical stability, it would prevent OLS from generalizing to unseen data. We will do the Singular Value Decomposition (S.V.D) of X : $X = U \Sigma V$ then $(X^T X)^{-1} = (V \Sigma^2 V^T)^{-1} =$,
 $(X^T X)^{-1}$ will have single value that are the squared inverse of the singular value of X . It will lead to extremely large singular value when X 's singular value is close to 0. It is a huge problem of numerical stable problems. We can fix it by Ridge Regression.



Ideas 4. Ridge Regression

Ridge Regression add a penalty on the size of coefficients into O.L.S. It will help us mitigate the issues of multicollinearity and overfitting.

Theorem 36. Ridge Regression

Ridge Regression penalize the entries of w from becoming too large. We can do this by adding a penalty term constraining the norm of w . For a fixed, small scalar $\lambda > 0$, we now have: $\min \|Xw - y\|_2^2 + \lambda \|w\|_2^2$ then $W_{RIDGE}^* = (X^T X + \lambda I)^{-1} X^T y$. We will chose a large enough hyperparameter λ to prevent the singular value of $(X^T X)^{-1}$ being close to zero.

Proof. We will show that $W_{RIDGE}^* = (X^T X + \lambda I)^{-1} X^T y$ is true and it achieve global minimum.

$$L(w) = \|Xw - y\|_2^2 + \lambda \|w\|_2^2 = w^T X^T X w - 2w^T X^T y + y^T y + \lambda w^T w.$$

$$\nabla_w L(w) = 0 \implies 2X^T X w - 2X^T y + 2\lambda w = 0 \implies W_{RIDGE}^* = (X^T X + \lambda I)^{-1} X^T y.$$

We will show that it is strongly convex and we compute Hessian again and determine it as positive-definite:

$$\forall w > 0, w^T (X^T X + \lambda I) w = \|Xw\|_2^2 + \lambda \|w\|_2^2.$$

Note that $(X^T X + \lambda I) = V \begin{pmatrix} \Sigma_2^T + \lambda I & 0 \\ 0 & \frac{1}{\lambda} I \end{pmatrix}$ It is full rank, invertible now, as the largest singular value is confirmed to $\frac{1}{\lambda}$. ■

Example 3.**Code 2.****Ideas 5. Featured Engineering**

The true function $y = f(x)$ maybe like a non-linear function but we can still turn it into linear least-squares, by adding a new features $\phi(x)$.

Theorem 37. Featured Engineering

We can devise some new function $\phi : \mathbb{R}^l \rightarrow \mathbb{R}^d$, called a feature map, that maps each raw data point $x \in \mathbb{R}^l$ into a vector of feature $\phi(x)$. The hypothesis function is $h_w(x) = \sum_{j=1}^d w_j \phi_j(x) = w^T \phi(x)$. Then the model is still linear with respect to the feature. The component function ϕ_j are called basis function, we can still use least-square to estimate the weight w . Therefore we replace the original data matrix $X \in \mathbb{R}^{n \times l}$ by $\phi \in \mathbb{R}^{n \times d}$, which has $\phi(x_i)^T$ as its i th row:

$$\min_w \|\phi w - y\|_2^2.$$

Example 4.**Ideas 6. Hyperparametal Finetuning**

We should do a lot of trial-and-error to find the best hyperparameter λ . When we do validation training, we will try a lot of hyperparameter λ and then training, return the parameter w^* . If the validation is bad and get a lot of errors, then do it again. If not, then return the best hyperparameter and parameter to do the test. Therefore we need to do a lot of validation.

Definition 70. K-Fold Cross Validation

We need to separate the validation data and training data. Cross-Validation is an alternative to having a dedicated validation set.

- Shuffle the data and partition it into k equally-sized blocks.
- For $i = 1, \dots, k$, train the model on all data except block i and evaluate the model using block i
- Average the k validation errors, this is our final estimate of the true error.

Example 5.**Code 3.**

0.29.4 Regression.2

Ideas 7. Is l^2 norm a good choice to estimate the model

We use l^2 norm to measure the error of our predictions and to penalize the model parameter, as it is a great design choice in statistical interpretations of regression, such as Gaussians, M.L.E and M.A.P.

Definition 71. Maximum Likelihood Estimation (M.L.E)

M.L.E aims to find the hypothesis model that maximizes the probability of the data.

Theorem 38. O.L.S satisfy M.L.E

The Maximum Likelihood Estimation is exactly same as Ordinary Least Square.

Proof. If we parameterize the set of hypothesis models with θ , we can express the problem as $\theta_{MLE} = \arg_{\theta} \max L(\theta; D) = p(\text{data}=D | \text{true model} = h_{\theta})$. The quantity $L(\theta)$ that we are maximizing is also known as the likelihood. Considering log we still working the same problem, because logarithms are monotonic functions. We have $P(A) < P(B) \iff \log P(A) < \log P(B)$. Let decompose the log likelihood :

$$l(\theta; X, y) = \log p(y_1, \dots, y_n | x_1, \dots, x_n, \theta) = \log(\prod_{i=1}^n p(y_i | x_i, \theta)) = \sum_{i=1}^n \log[p(y_i | x_i, \theta)]$$

Let $p(y_i | x_i, \theta)$ from a Gaussian $Y_i | \theta \approx N(h_{\theta}(x_i), \sigma^2)$.

Then we will have $\theta_{MLE} = \arg_{\theta} \max l(\theta; X, y) = \arg_{\theta} \max_{\theta} - (\sum_{i=1}^n \frac{(y_i - h_{\theta}(x_i))^2}{2\sigma^2}) - n \log(\sqrt{2\pi}\sigma) = \arg_{\theta} \min \sum_{i=1}^n (y_i - x_i^T \theta)^2$

Therefore the M.L.S is just O.L.S problem! ■

Example 6.

Code 4.

Definition 72. Maximum a Posteriori

Maximum a Posteriori Estimation aims to find the model for which maximizes the probability of the model $\theta_{MAP} = p(\text{true model} = h_{\theta} | \text{data} = D)$ which is a posterior.

Theorem 39. Ridge regression satisfy M.A.P

Proof. Using the Bayes Rules, we have $\theta_{MAP} = \arg_{\theta} \max P(\text{true model} = h_{\theta} | \text{data} = D)$

$$= \arg_{\theta} \max \frac{P(\text{data}=D | \text{truemodel}=h_{\theta}) P(\text{truemodel}=h_{\theta})}{P(\text{data}=D)} = \arg_{\theta} \max \log P(\text{data} = D | \text{truemodel} = h_{\theta}) + \log P(\text{truemodel} = h_{\theta})$$

$$= \arg_{\theta} \min - \log P(\text{data} = D | \text{truemodel} = h_{\theta}) - \log P(\text{truemodel} = h_{\theta})$$

$$\text{then } \theta \approx N(\theta, \sigma^2) \text{ such that } \theta_{MAX} = \arg_{\theta} \min (\sum_{i=1}^n (y_i - h_{\theta}(x_i))^2 + \frac{\sigma^2}{\sigma_h^2} \sum_{j=1}^2 \theta^2$$

This is just the a Ridge Regression! ■

Code 5.

Example 7.

Ideas 8. M.L.E vs M.A.P

Note that M.A.P is just like M.A.P except we add a term $P(\text{truemodel} = h_{\theta})$, this term is prior over our true model. However what parameters will M.L.E and M.A.P select given any data set D ?

Considering M.L.E, it will choose the parameter θ containing maximum likelihood but it might have high variance. A slightly different data set will alter the predicted model

M.A.P aim to balance the variance and likelihood. If the model weight should be small then the prior will centered at zero with small variance, M.A.P will choose the parameter maximizes the posterior probability



Definition 73. *Bias-Variance Tradeoff*

$\varepsilon(x, h) = (E[h(x; D) - f(x)])^2 + \text{Var}(h(x; D)) + \text{Var}(Z)$ is the bias -variance to measure the effectiveness of hypothesis.

Proof. Let $Y = f(x) + Z$ be a noisy measurement of the true function $f(x)$, we might try to use M.A.P and M.L.E to recover it but how to measure the hypothesis effectiveness? We will form a theoretical metric to measure the effectiveness of hypothesis function $h(x)$, the expected error between the hypothesis and the observation $Y = f(x) + Z$ such that:

$$\varepsilon(x, h) = \mathbb{E}[(h(x; D) - Y)^2] = \dots = (E[h(x; D) - f(x)])^2 + \text{Var}(h(x; D)) + \text{Var}(Z)$$

where it separate as the *bias*² of the method and variance of the method as well as irreducible error. ■

Code 6.**Theorem 40.** *Matching Pursuit Algorithm*

Initialize the weight $w^0 = 0$ and the residual $r^0 = y - Xw^0 = y$
 while $\|w\|_0 < k$ do
 {
 Find the feature i for which the length of the projected residual onto x_i is maximized
 $i = \arg \min(\min_v \|r^{t-1} - vx_j\|) = \arg \max \frac{|\langle r^{t-1}, x_j \rangle|}{\|x_j\|}$
 update the i th feature entry of the weight vector
 $w_i^t = w_i^{t-1} + \frac{|\langle r^{t-1}, x_j \rangle|}{\|x_j\|}$ update the residual $r^t = y - Xw^t$
 }

Code 7.**Example 8.**

0.29.5 Dimensionality Reduction

Theorem 41. *Principal Component Analysis*

Proof.

Code 8.

Example 9.

Theorem 42. *Canonical Correlation Analysis*

Proof.

Code 9.

Example 10.



0.30 Big Data and its Computation



0.31 Mathematics of Machine Learning



0.32 Operation Research



0.33 Optimization Research



0.34 Game Theory



0.35 Formal Language and Automata Theoeoy



0.36 Algorithm and mathematics



0.37 Computational Complexity Theory



0.38 Coding Theory and Cryptography



0.39 Mathematics fo Physics



0.40 Mathematics for Biology



0.41 Mathematics for Chemistry



0.42 Mathematics for Engineer



0.43 Honour and Credits

Credit 1. *MATH JESSUS, CHAIRMAN, NEW TA Kelvin*

He inspires every math major friends when he proof all stuff in undergraduate mathematics.

Credit 2. *ALGEBRA TA Andy*

He uses algebra to brainstorm every math major students and he is the BEST TA.

Credit 3. *WEIRD BUT KIND MATH MAJOR FRIEND Wiki*

He is one of my best friend, who always hang out with me when my midterm and final result..... look "nice"...

Credit 4. *BREAVEST MATH MAJOR Alex*

He is the bravest math major in the world as he didn't study M1 or M2 in HKDSE.BUT still want to be a MATH MAJOR in CUHK. Also we finished a lot of CODING projects all night long.

Credit 5. *HALF RISK MANAGEMENT, PURE BLOOD MATH MAJOR Yuki*

She is crazy as she want to double major in both mathematics and risk management.She also inspired everyone to contribute to the society.

Credit 6. *BEST ACADEMIC ADVISER Dr.Cheung Leung Fu*

He inspired me to minor CS then I can have a "good-looking" GPA script. Also he is really kind-hearted, spending a lot of time to hang out with students. He also teach me how to borrow math books in CUHK.

Credit 7. *THE IDEAL MODEL,CUHK MATH ICON,THE MOST RIGIOUS Dr Fong Wing Chung*

...Sorry sir, I still studying math right now. Please forgive my poor English and not Rigious Proof. However, you let me know I' maybe not suitable studying MATH major. THANK YOU!

Credit 8. *CHEM MAJOR but GOT SCAM by CHEM DEPARTMENT and TAKE MATH3020 Jojo*

Thank you for hanging with us to go through Axiomatic Set Theory.You are insane!

Credit 9. *Data Analysis Best Teammate and my friend Sammy Wu Hong Ru*

He save me from many mini-project deadline, help me a lot.



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